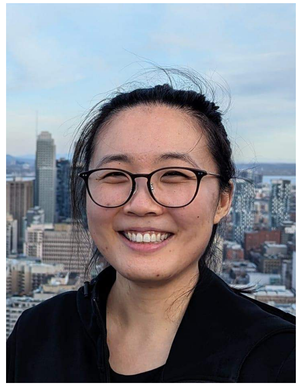
A new primitive for uncertainty quantification

Hong Namkoong, Oct 2026


Tiffany Cai

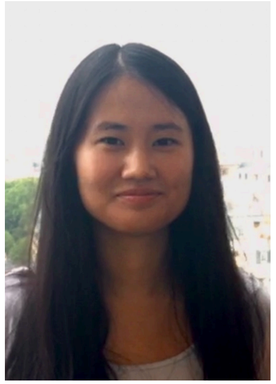

Daksh Mittal


Dan Russo


Jimmy Wang


Naimeng Ye


Rich Zemel


Kelly Zhang


Tom Zollo

Progress in AI

- 30% of patients develop atrial fibrillation after open heart surgery
- Risk score can allow surgeons to intervene, e.g., clip atrial appendage

Approach 1: Standard ML

Work hard to train a fancy model & carefully tune it

Approach 2: Lazy prompting

Prompt a pre-trained transformer with your data and x, ask it to predict y



Hollman et al. ‘25

Progress in AI

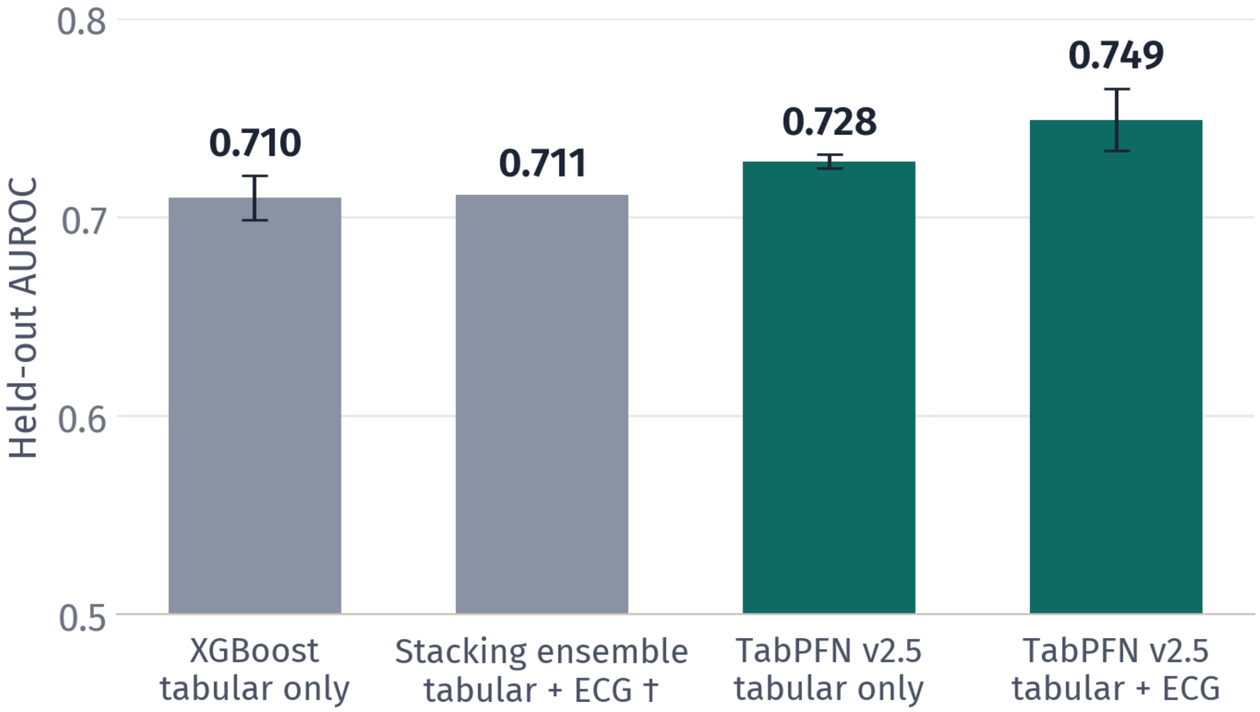
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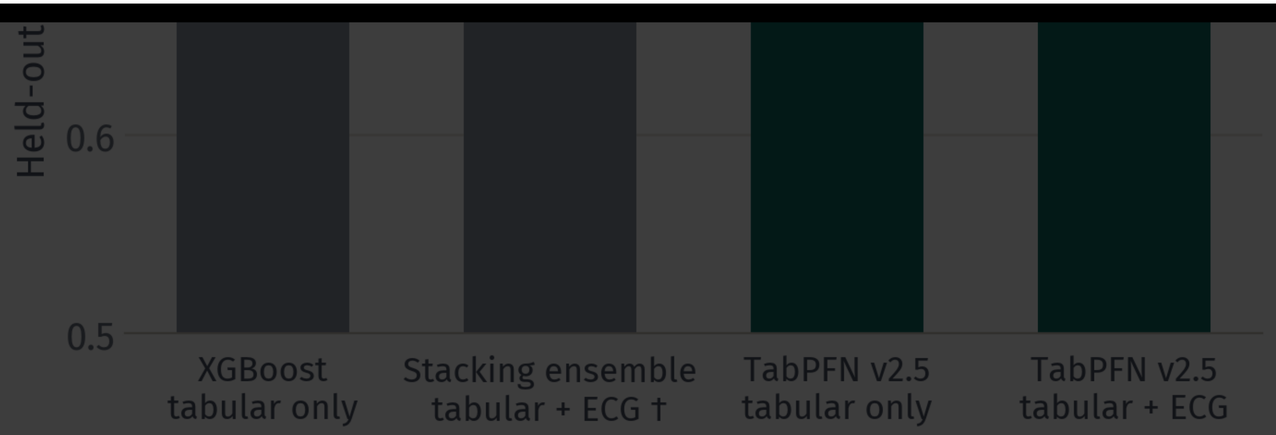
Progress in AI

- 30% of patients develop atrial fibrillation after open heart surgery
- Risk score can allow surgeons to intervene, e.g., clip atrial appendage

Premise of this talk: AI can i) seamlessly integrate data modalities
ii) transport well under distribution shift

Approach 2: Lazy prompting

Prompt a pre-trained transformer with your data and x, ask it to predict y



Knowing what you don’t know

- Several lines of work tackle uncertainty quantification
  - Bayesian neural networks, GPs, ensembles, epistemic neural nets, conformal prediction, multi-calibration…many other interesting ideas
- But these ideas have not “scaled” to the level of LLMs

Knowing what you don’t know

- Several lines of work tackle uncertainty quantification
  - Bayesian neural networks, GPs, ensembles, epistemic neural nets, conformal prediction, multi-calibration…many other interesting ideas
- But these ideas have not “scaled” to the level of LLMs

- Why? The bitter lesson: only thing that works is scaling up training data by

  1. optimizing loss on web-scale training data
  2. testing engineering innovations based on val loss

Example: adaptive student assessment

Why probabilistic modeling is hard

X's exogenous & plentiful



Amin

Z = prior info

Amin generally solves single-digit division problems fluently and uses repeated subtraction when working with larger numbers. When asked to explain his reasoning, he often relies on 'what feels right' rather than written strategies. He's more confident when problems are framed in real-life contexts, like sharing food or money, and shows persistence even when confused.

X_1 = A juice bottle has 3.9 liters. If I share it equally among 13 friends how many liters does each get?"

Y_1 = I divided 3.9 by 13 and got 3, so I think each friend gets 3 liters. Answer: D

X_2 = If 13 aunts each give Timmy $0.75 for Christmas, how much does he have?

Y_2 = I multiply 13 × 0.75 and get $9.75 total. Answer: B

Which questions X will Amin struggle with?

Example: adaptive student assessment

Why probabilistic modeling is hard

- For “environment Z”, we observe $Y_t = f^\star(X_t) + \varepsilon_t$ for t=1,…,10
- Goal of probabilistic inference: posterior distribution over $f^\star(\,\cdot\,)$
- But how do we draw a good confidence interval around a function whose values are never observed?



Why typical ML is useless here

- Hard to fit Y|X with a few samples
- This only gives you a single $\hat{f}(x)$
- Unclear how to leverage Z

Example: adaptive student assessment

Why probabilistic modeling is hard

- Statisticians have to resort to not using any ML:

Z = prior info



Amin

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Prior $f^\star(\cdot) \sim \mathbb{P}_{f^\star}(\cdot \mid Z)$

Likelihood $\varepsilon \sim \mathbb{P}_{\varepsilon}(\cdot \mid Z)$

- This is only easy when $Z \in \mathbb{R}^{10}$ and $f^\star(x) = \beta^{\star T} x$
- Experts spend year(s) comping up with a reasonable proxy ("modeling")

Why is this so hard?

- Both $f^\star(\,\cdot\,)$, ε are never observed
- How are we supposed to benchmark models if we never see them?

Latent parameter has no physical meaning!

Hard to check whether your unicorn is better than mine



A new primitive for uncertainty quantification

Uncertainty comes from **missing data** yet to be observed

$$\mathbb{P}(Y_{t+1:T} \mid \underbrace{Z, X_{1:T}, Y_{1:t}}) = \int \prod_{s=t+1}^{T} \underbrace{\mathbb{P}(Y_s \mid X_s, f^\star, Z)} \; d\,\underbrace{\mathbb{P}(f^\star \mid Z, X_{1:t}, Y_{1:t})}$$

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future — past — likelihood (DGP) — prior

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future | past | likelihood (DGP) | prior

- **Classical Bayesians** use likelihoods and priors without physical meaning
- A **predictive view** takes *one-step predictions* as the modeling primitive

$$\mathbb{P}(Y_{t+1:T} \mid Z, X_{1:T}, Y_{1:t}) = \prod_{s=t}^{T-1} \underbrace{\mathbb{P}(Y_{s+1} \mid Z, X_{1:s+1}, Y_{1:s})}_{\text{one-step prediction}}$$

Foundation models

Preliminaries

- Observation: we have transformers $\hat{p}(Y_{t+1} \mid Z, X_{1:t+1}, Y_{1:t})$ **meta-learned** across many semi-synthetic datasets



Foundation models

Preliminaries

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Prior-Labs nvidia/Kumo Tabular LimiX TabFM

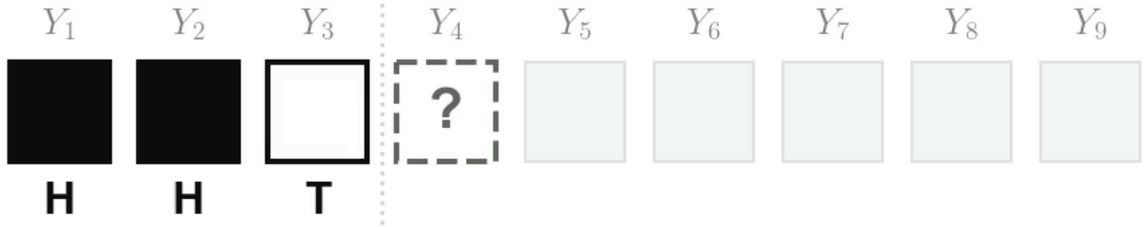
- BayesML proposes repeatedly drawing from this distribution
[Muller et al., ('24), Nguyen & Grover ('23)]

- Problem: this gives you **predictive intervals**, but not a distribution over $f^\star(\cdot)$! Need to separate out ε.

Naive way: IID

every flip uses the prediction from the 3 observed flips

A coin with unknown chance of heads. Observed: H, H, T.



$$\mathbb{P}(Y_4 = \mathrm{H} \mid Y_{1:3}) = \frac{1+2}{2+3} = 0.60$$

chance of heads for the next flip

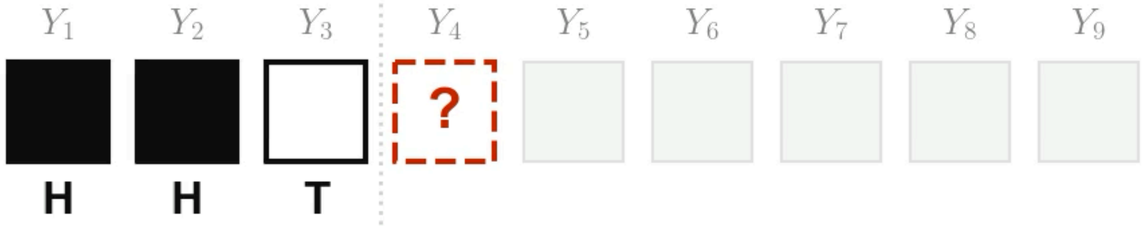
Three flips observed. Predict the next flip from them.

Our autoregressive approach

Correct way: autoregressive

each new flip goes back into the context

Same coin, same three observed flips.



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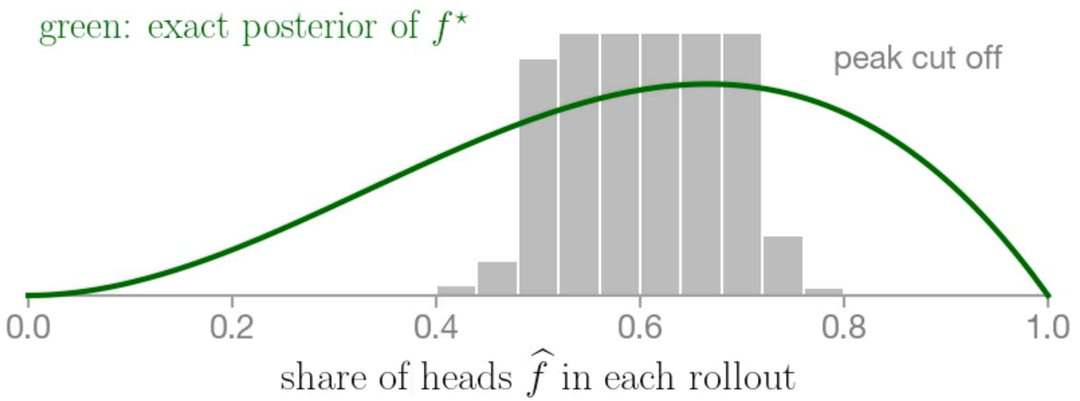
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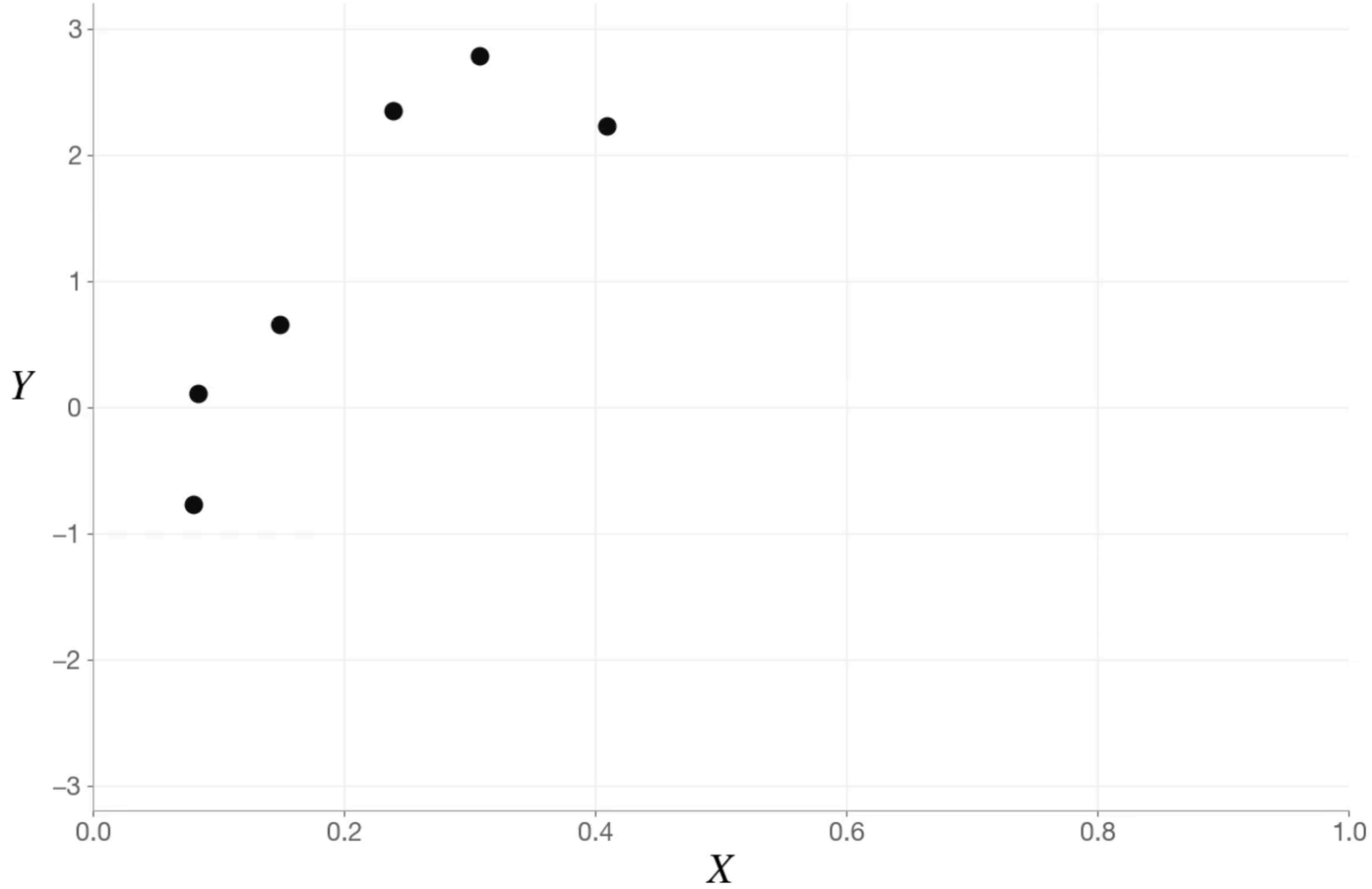




Every rollout ends near 0.60. The spread is coin-flip noise, far narrower than the posterior.

Contexts X and heteroskedasticity

Correct way: autoregressive generation each generated answer goes back into the context



Context

s	X_s	Y_s
1	0.08	-0.77
2	0.08	0.11
3	0.15	0.65
4	0.24	2.35
5	0.31	2.78
6	0.41	2.23

Observed so far: $X_{1:t}$, $Y_{1:t}$ with $t = 6$

Language data

Prompt

Z	Amin solves single-digit division fluently and relies on ‘what feels right’ rather than written steps.
X_1	A juice bottle has 3.9 liters. If 13 friends share it equally, how much does each get?
Y_1	I divided 3.9 by 13. 13 goes into 39 three times, so each friend gets 3 liters.
X_2	If 13 aunts each give Timmy $0.75, how much money does he have?
Y_2	Ten aunts give $7.50 and three more give $2.25, so Timmy has $9.75.

- Observed: background Z and two answers

“Modeling” = Training sequence predictors

	Classical Bayesian modeling Uncertainty comes from unknown $f^\star$ and ε	**This work: predictive view** Uncertainty comes from missing future data
Ingredients	Prior $f^\star(\cdot) \sim \mathbb{P}_{f^\star}(\cdot\,|Z)$ Likelihood $\varepsilon \sim \mathbb{P}_{\varepsilon}(\cdot\,|Z)$	Autoregressive probabilities $p(Y_{t+1}|\mathrm{Z}, X_{1:t+1}, Y_{1:t})$
Model of uncertainty	$\mathbb{P}(f^\star \in \cdot\,|Z, X_{1:t}, Y_{1:t})$	$p(Y_{t+1:T}|Z, X_{1:T}, Y_{1:t})$
Updating beliefs	Posterior inference, e.g., MCMC	In-context learning: append (X_t, Y_t) to the prompt
Checking the model	Not possible: $f^\star$ and ε are never observed	Sequence log loss $\ell(p)$ on held-out students

Predictive loss controls posterior quality

- Sequence prediction loss $\ell_{t:T}(p) := -\,\mathbb{E}_t\left[\sum_{s=t}^{T-1} \log p(Y_{s+1} \mid Z, X_{1:s+1}, Y_{1:s})\right]$

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(Ye & **N.** ’26) As long as there is $\hat{f}$ fitted from data generated by $\hat{p}$

$$\left|\underbrace{\mathbb{P}(\tau(f^\star) \in \widehat{C}_t \mid \text{past}) - 95\,\%}_{\text{error in confidence interval coverage}}\right| \le \sqrt{1 - \exp\left(-\lim_{T\to\infty}\left(\ell_{t:T}(\hat{p}) - \ell_{t:T}(p^\star)\right)\right)}$$

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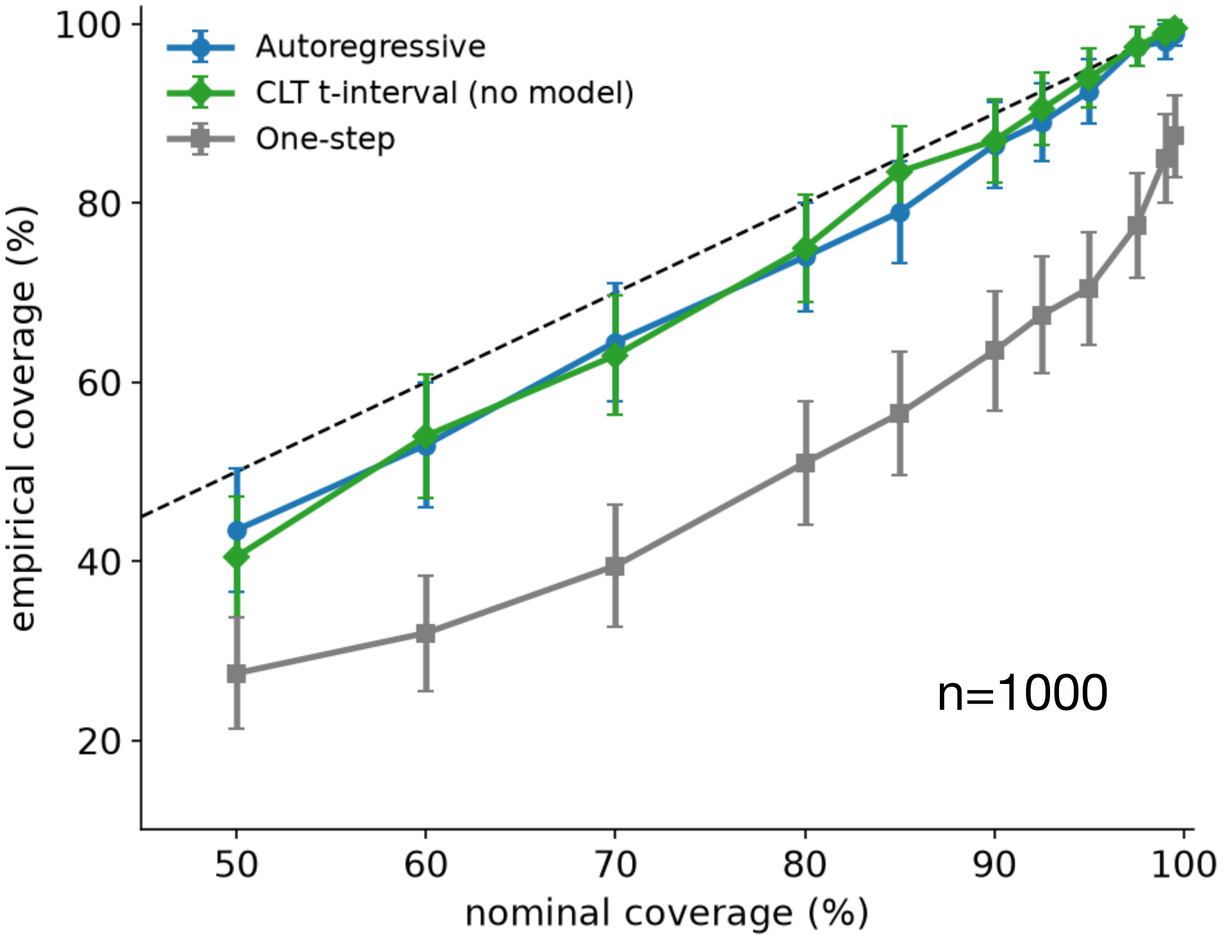
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- Predictive loss over **right outcomes** controls quality of UQ!
- $\hat{p}$ doesn't have to be exchangeable so long as it provides a consistent estimator

Sanity check

Can we turn a tabular foundation model into a statistician?

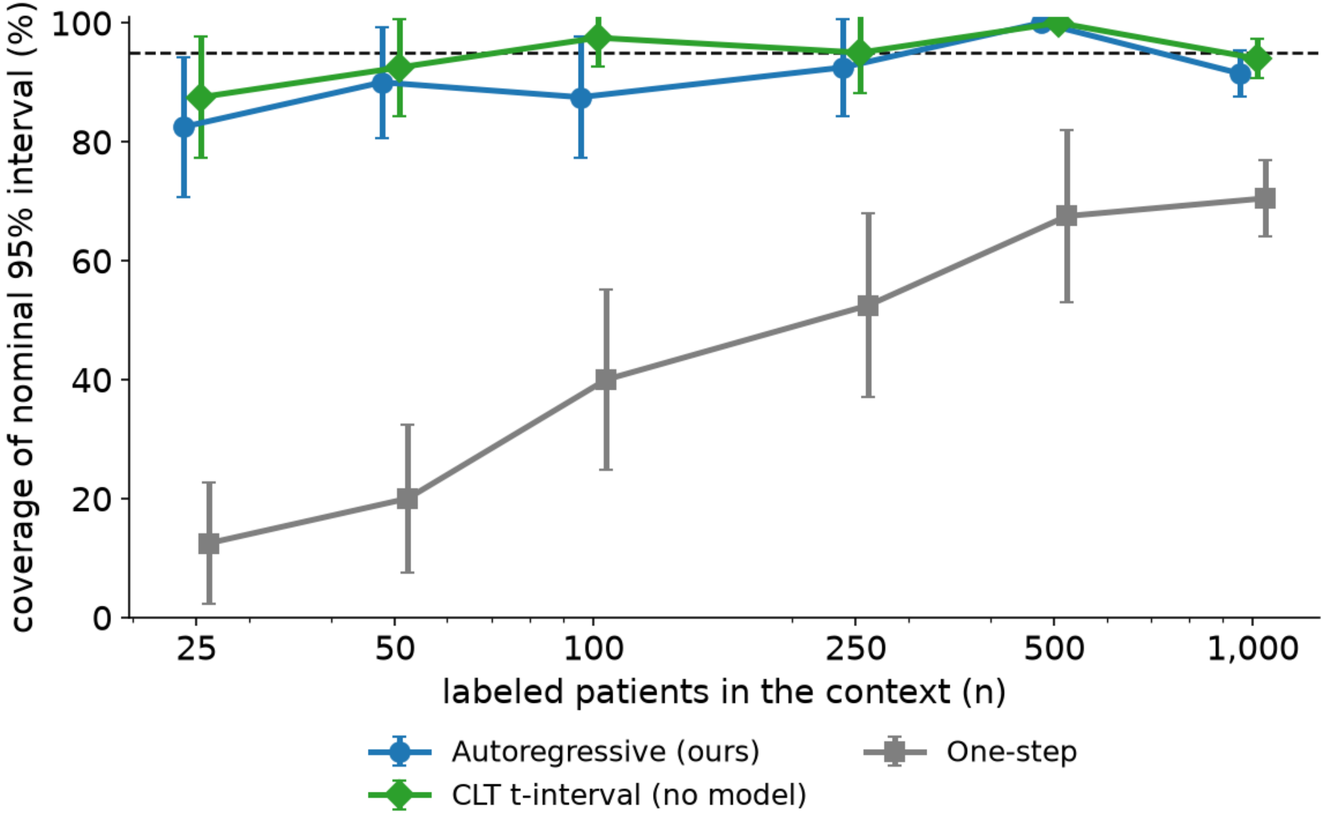
- Let's consider a **one-dimensional** estimand instead of a whole function $f^\star$
- In this case, classical statistical machinery (aka CLT) works well
- Estimand: prob. of prolonged ventilation at CUMC
- Can tabular FMs produce good confidence intervals?



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Comparison to industry standard

Fit $\hat{f}(x)$ on bootstrapped data, and pretend $\hat{f}_1, \ldots, \hat{f}_{20}$ is your universe

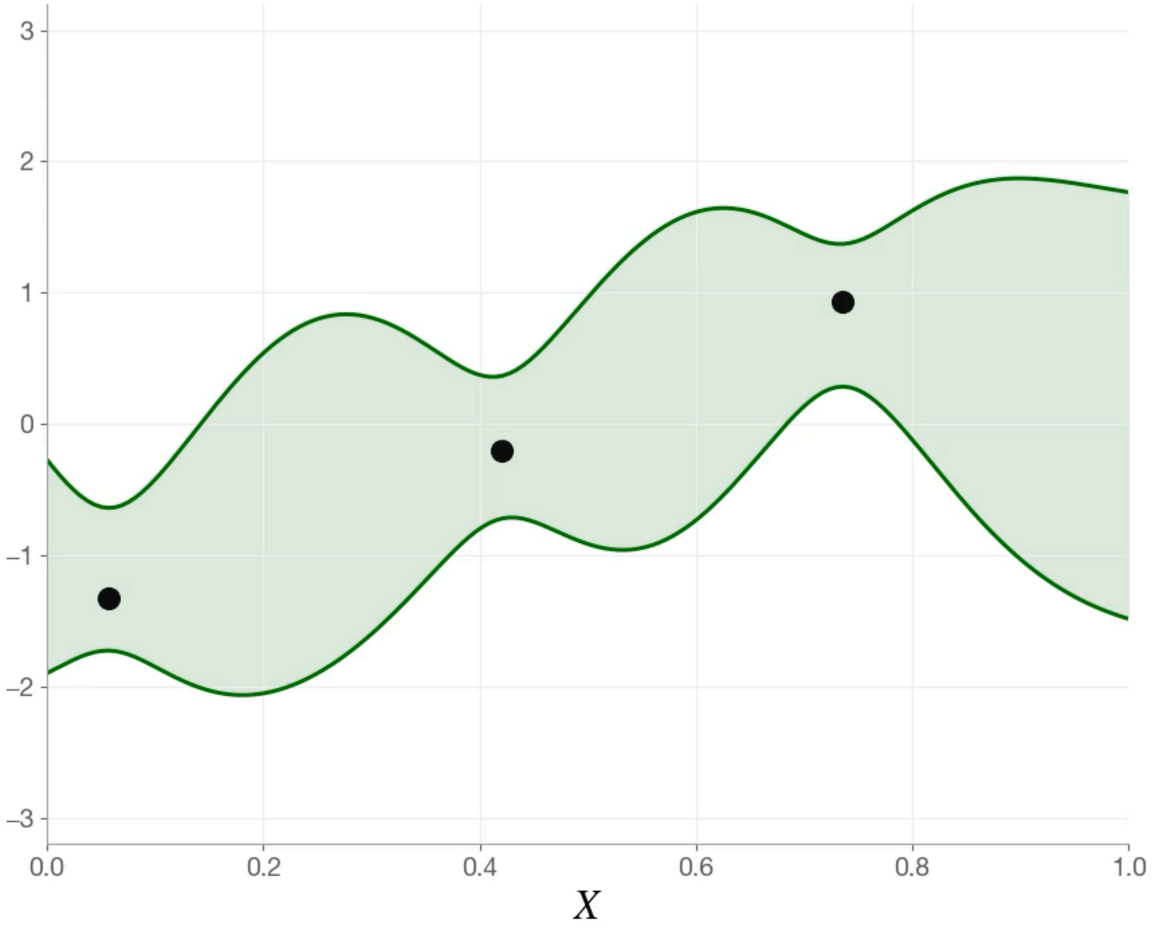
Challenges

- Very unstable: we need to regularize/smooth when data is small, but it's unclear how to because there is no notion of val loss to tune
- Hard to update when you get one more response (X_{11}, Y_{11}) from Amin
- Can't learn across Z's ("prior information")
- **Consequently, cold start problem largely remains unsolved**

Bootstrap ensemble vs. ours with only 3 observed answers

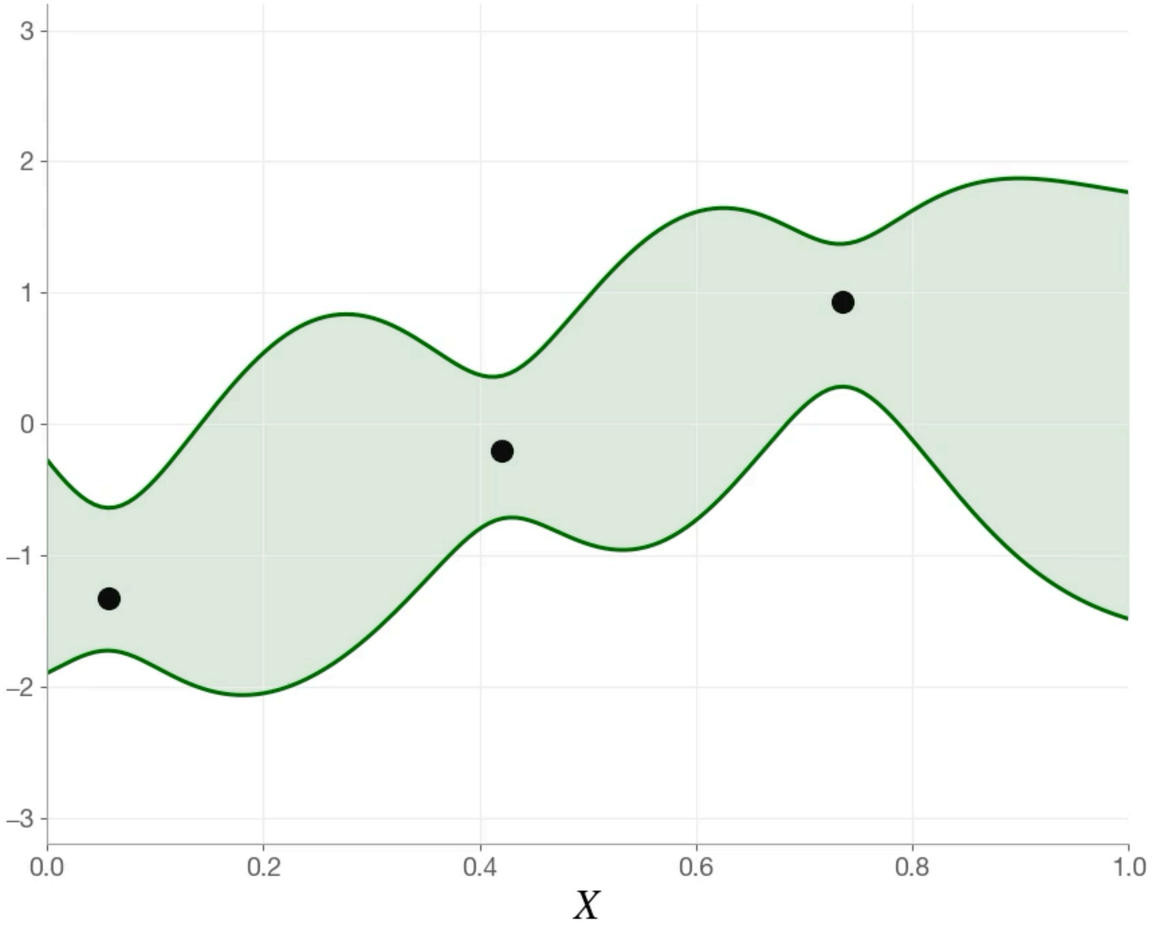
Bootstrap ensemble

Resample the 3 observed answers, refit



Ours

Generate the missing answers to T = 300, refit



Three observed answers. Green: the exact posterior of $f^\star$

Related work

Modeling joint distribution was challenging before LLMs

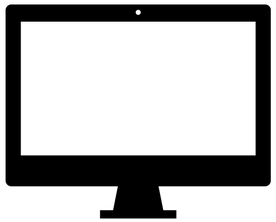
- De Finnetti (’29): exchangeable sequence modeling = latent factor models
  - Geisser (’71), Rubin (’78) espouse similar idea, but found it “too burdensome to be worthwhile”; resorts to typical latent variable modeling
- Martingale posteriors handpick autoregressive probabilities and don’t view sequence prediction loss as the fundamental learning objective
  [Berti et al. (’98, ’21, ’22), Fortini et al. (’14, ’23), Fong et al. (’23), Holmes & Walker (’23), Falck et al. (’24)]
- Our view: it doesn’t matter whether $\widehat{p}(\,\cdot\,)$ is exchangeable so long as it’s a good sequence predictor

This allows us to repurpose foundation models as a statistician!

How does this help decisions?

Vignette in adaptive experimentation

Adaptive experimentation as a MDP

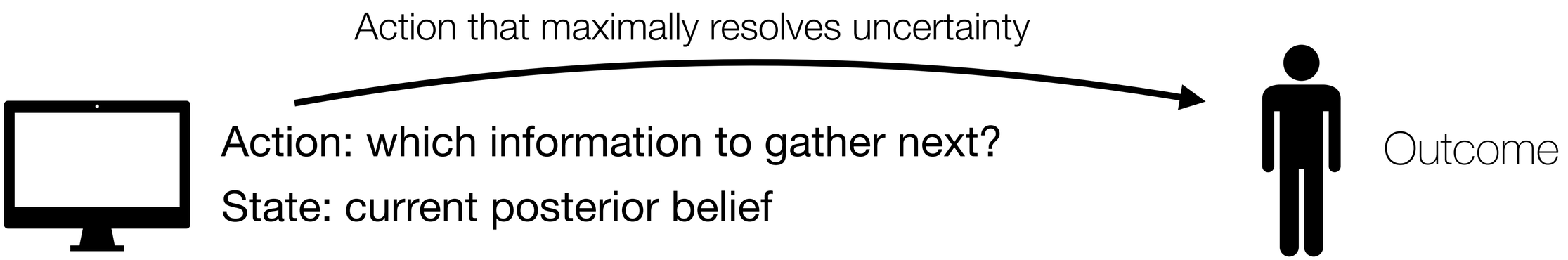


Action: which information to gather next?

State: current posterior belief

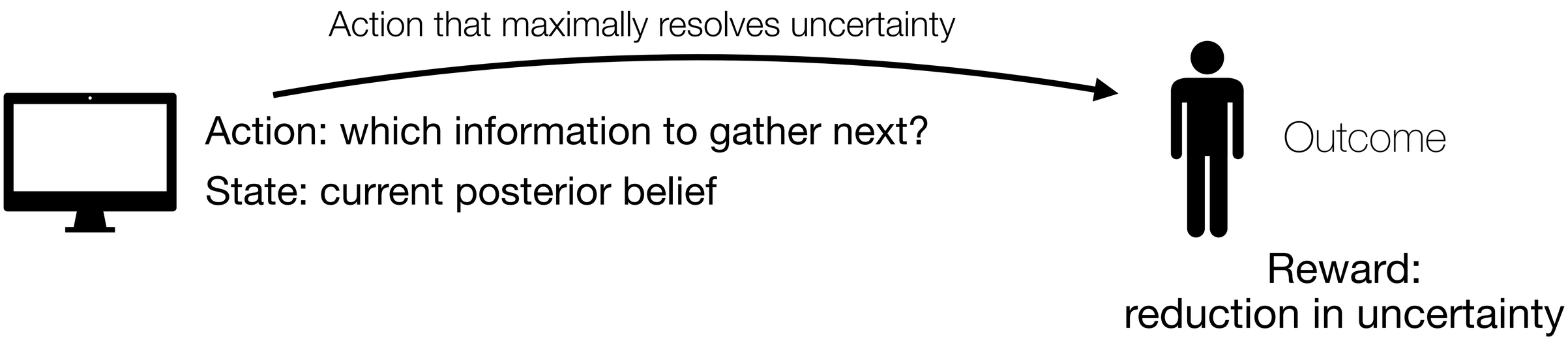
[Gittins & Jones (’74), Kaelbling et al. (’98), Ghavamzadeh et al. (’15), Frazier et al. (’09), Powell & Ryzhov (’12), Brown & Smith (’13), Wang & Zenios (2020)]

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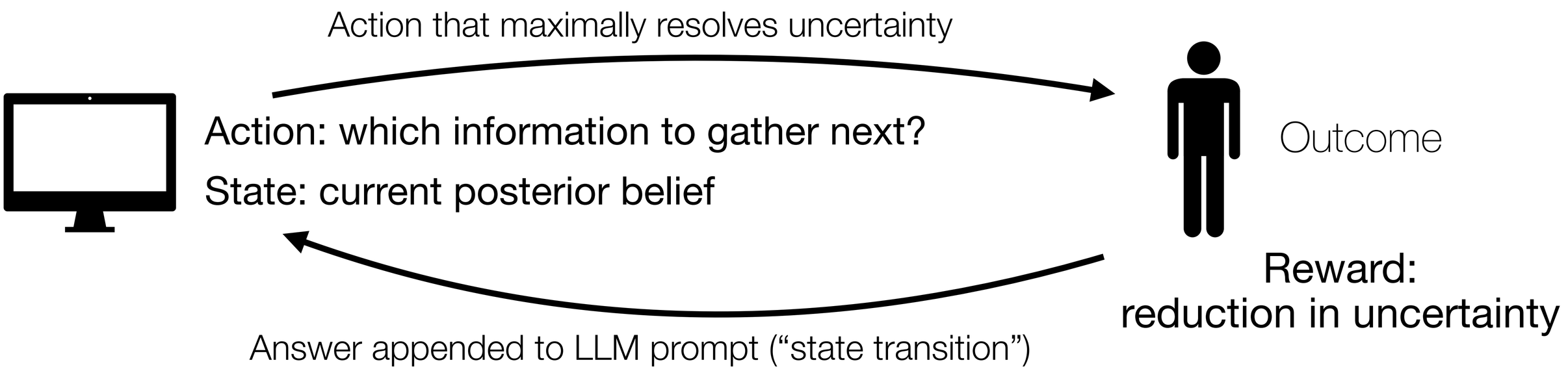
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Adaptive exams to predict prolonged ventilation

Retrospective case study

- Substantial heteroskedasticity over covariates: demographics, lab results, comorbidities, preop meds, op type etc.

Panel	What it contains	Cost	Ordered
Basic labs	creatinine, hematocrit, WBC, platelets, INR, albumin, hemoglobin	1	100%
HbA1c	HbA1c	1	62%
Echo	ejection fraction, LV end-diastolic diameter, LV systolic dimension	3	99%
Pulmonary function	FEV_1, DLCO	5	39%
PA pressure	systolic pulmonary artery pressure	8	41%

- Ignorability: clinician's orders independent of outcome given observed history. (This is incorrect, e.g., patients who got PFT are sicker)

Adaptive exams to predict prolonged ventilation

Retrospective case study

Action: HbA1c, Echo, PFT, PA pressure

State: TabPFN prompted with information so far



Exam

Adaptive exams to predict prolonged ventilation

Retrospective case study

Action that maximally resolves uncertainty

Action: HbA1c, Echo, PFT, PA pressure

State: TabPFN prompted with information so far

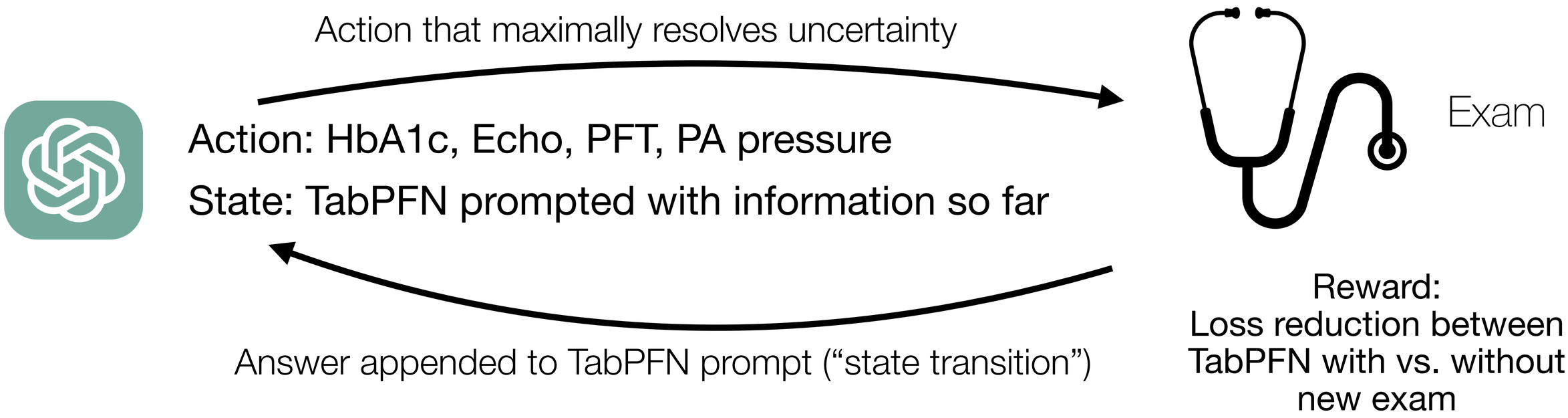


Exam

Reward:
Loss reduction between TabPFN with vs. without new exam

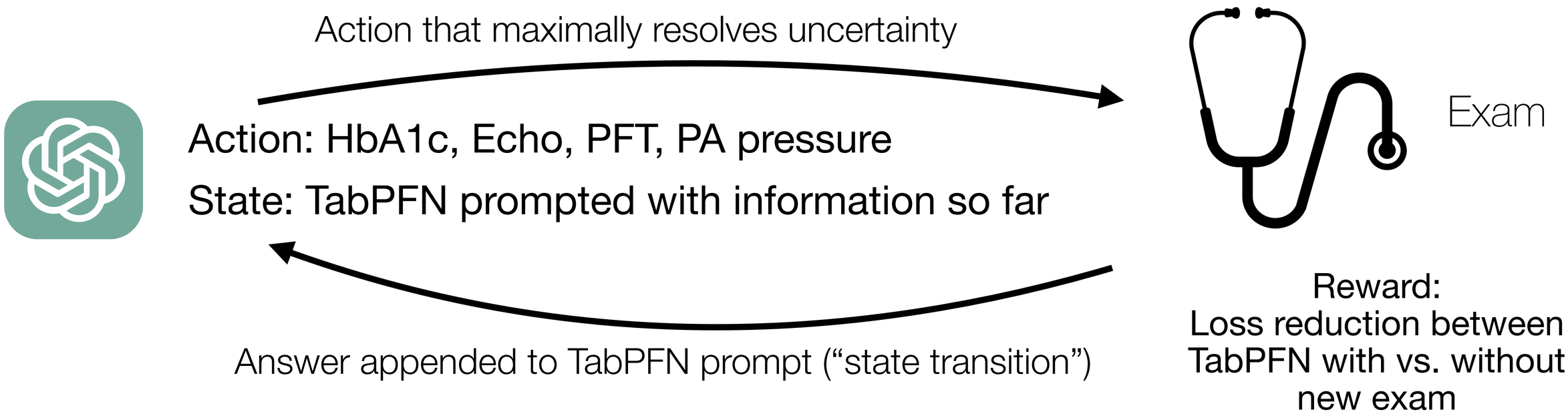
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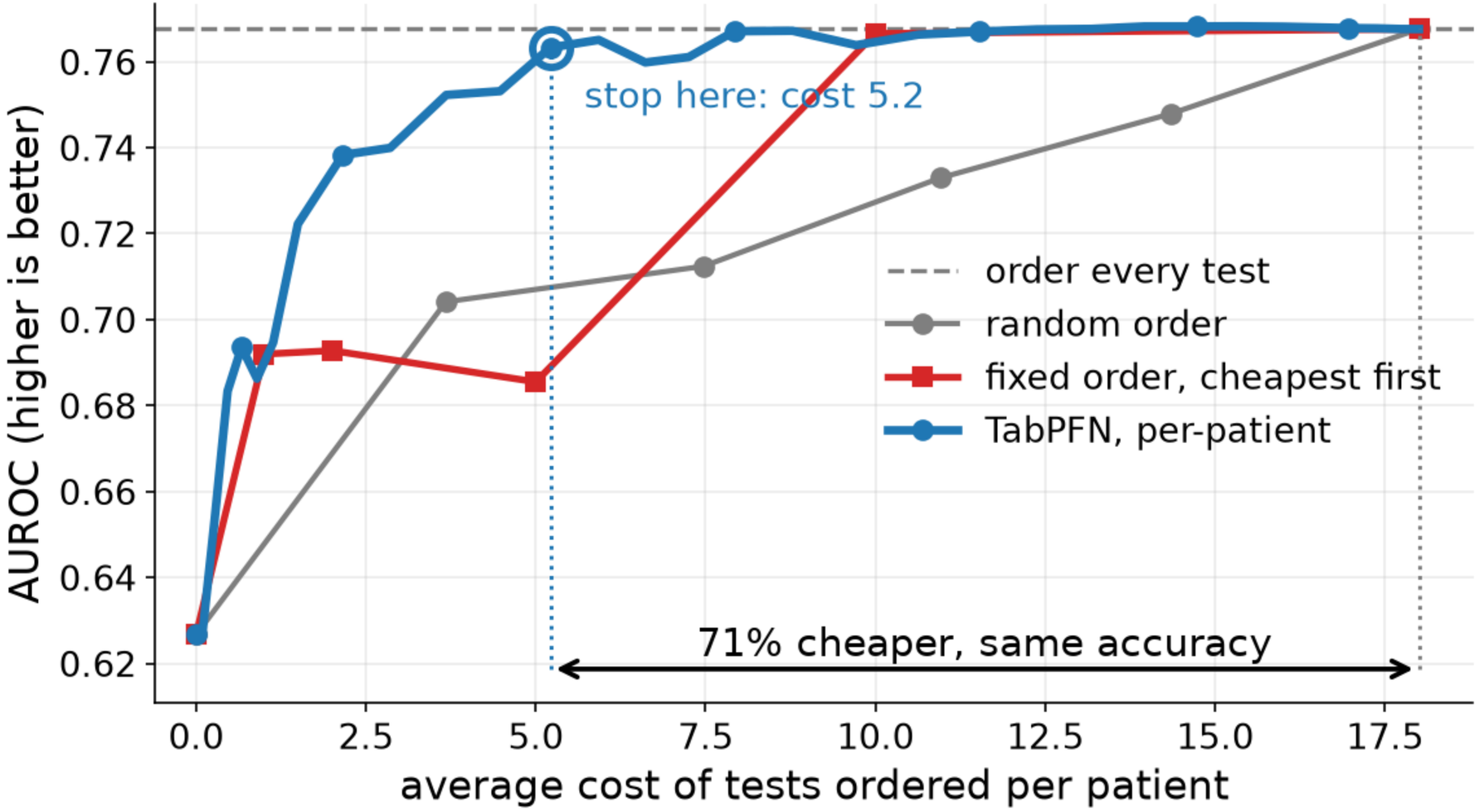
Retrospective case study



- Today: pick exams with highest ratio of uncertainty reduction / cost

Adaptive exams to predict prolonged ventilation

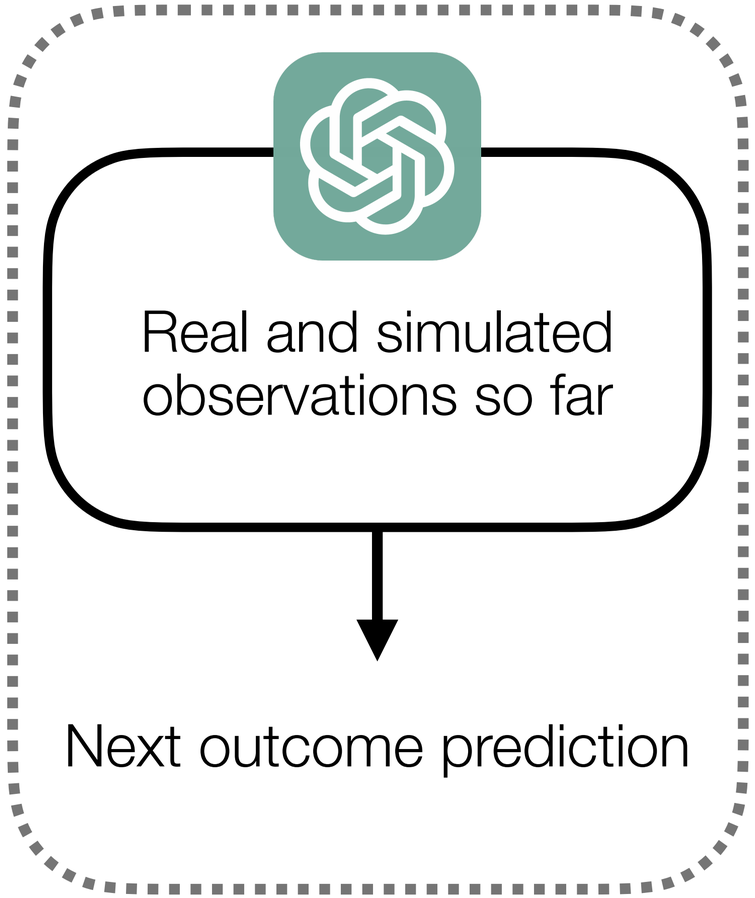
Retrospective case study



Conceptual advantages

- New information can be simulated by masking tokens in the tabular foundation model regardless of feature or outcome
- Value of new info measured by comparing the sequence model's prediction with & w/o
- Rollouts have known dynamics (in-context learning within a transformer)
- Allows auto-differentiation-based planning (aka model predictive control)

Sequence model serves both as a state, a simulator of the future, and prediction model



Natural language example

Real-world dataset from an online education platform

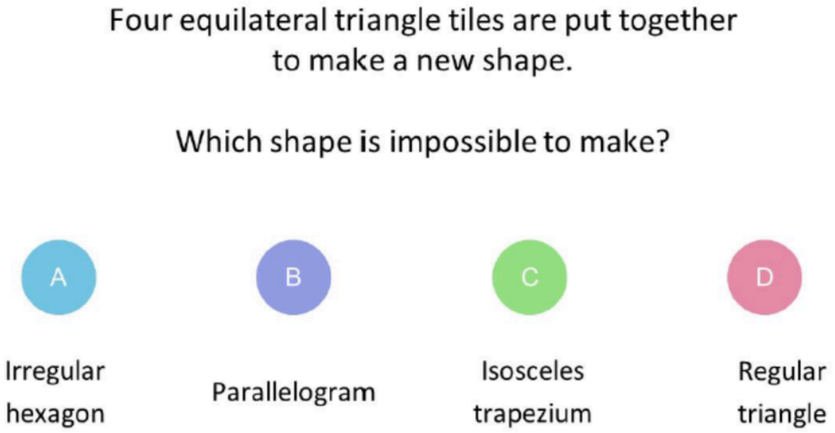
- ~1K multiple choice questions on algebra, number theory, and geometry
- Finetune Llama 3.1on individual responses from students (LoRA)

First question: uses both skills

3.9 ÷ 13 = ?

Student

Example question from EEDI

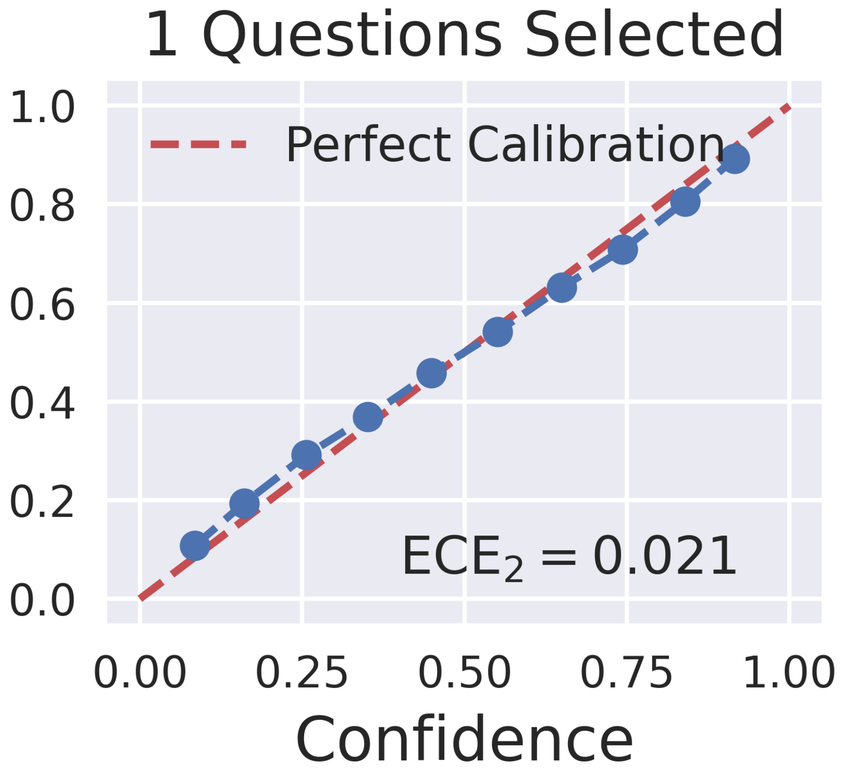


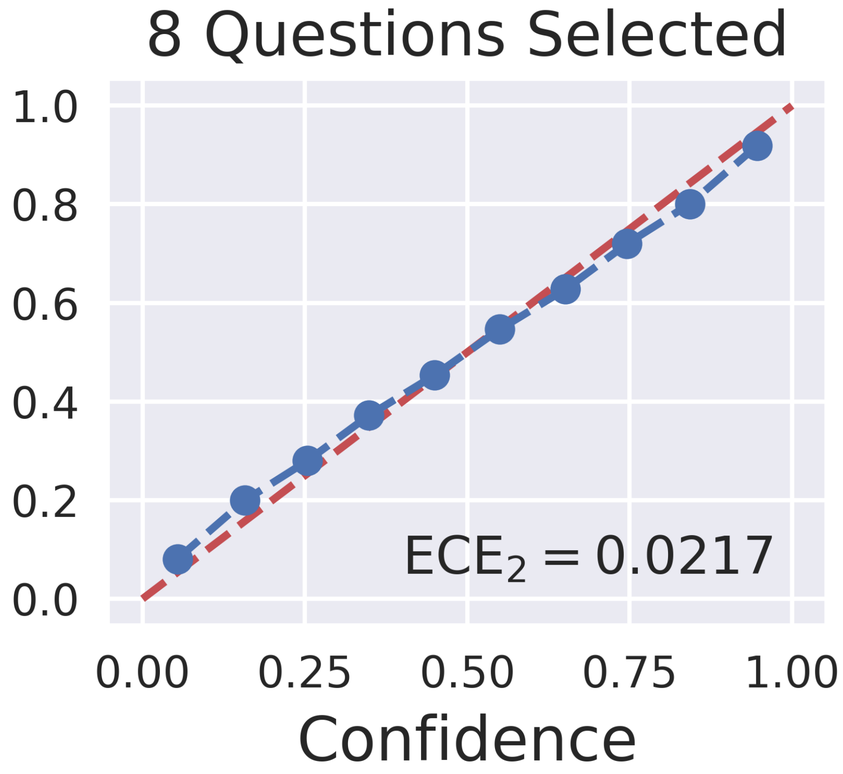
Diagnostic questions: The NeurIPS 2020 education challenge. https://diagnosticquestions.com/ Wang et al. (2020)

Natural language example

Predicted probabilities on student answer sensible?

y = Fraction(correct | confidence = x); we want this to be equal to x





Adaptive exams for student assessment

MDP formulation

Predictive view	Reduce uncertainty on future responses as measured by Entropy(student answers to all questions in problem bank)

Adaptive exams for student assessment

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Action: which question to pick next?

State: current posterior

Adaptive exams for student assessment

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Select Q that can maximally resolve future uncertainty



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Answer

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State: current posterior



Answer

Reward:
reduction in entropy

Adaptive exams for student assessment

Predictive view

Reduce uncertainty on future responses as measured by
Entropy(student answers to all questions in problem bank)

Select Q that can maximally resolve future uncertainty



Action: which question to pick next?

State: current posterior



Answer

Reward:
reduction in entropy

Answer appended to LLM prompt (“state transition”)

[Gittins & Jones (’74), Kaelbling et al. (’98), Ghavamzadeh et al. (’15), Frazier et al. (’09), Powell & Ryzhov (’12), Brown & Smith (’13), Wang & Zenios (2020)]

Adaptive exams for student assessment

Predictive view	Reduce uncertainty on future responses as measured by Entropy(student answers to all questions in problem bank)

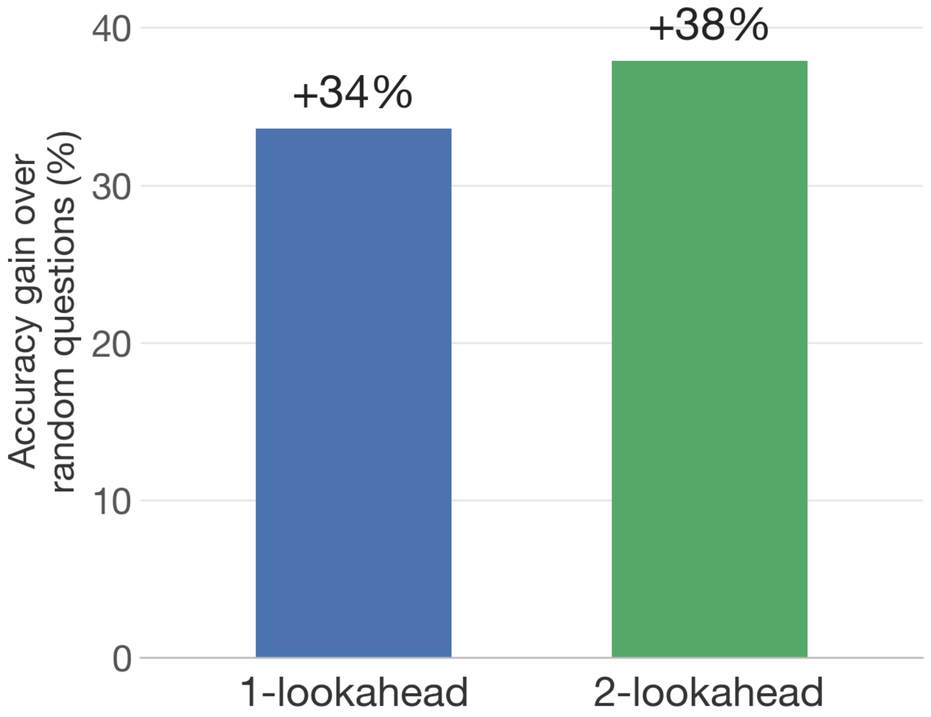
To (approximately) solve this MDP, we must

1. form informed prior based on information on the student
2. use question and solution to comprehend areas of largest uncertainty
3. select questions balancing exploration & exploitation
4. sharpen beliefs based on the student's answer

Adaptive exams for student assessment

Adaptive vs random questions

- Most of a student's answers match what most students say, so the population prior already predicts them
- What we need to learn is where this student differs, e.g., Amin divides well but misplaces decimals. Each student is scored on answers that fewer than 30% of students give.
- **There, adaptive questions predict the student up to 38% more accurately than random questions**



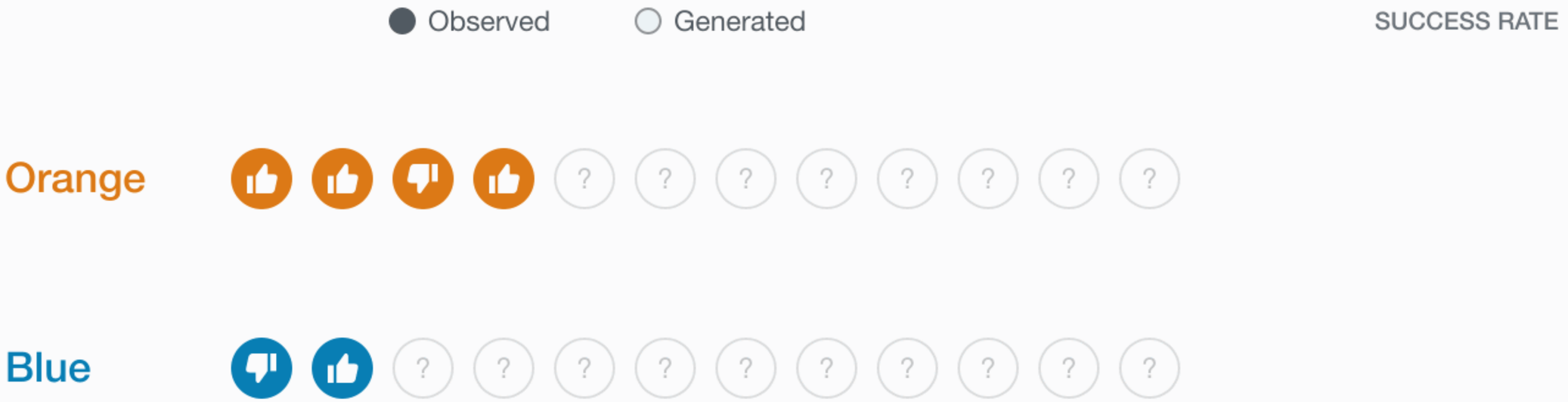
Theoretical insights

Progress in LLM pre-training directly improves interactive decision-making performance

[Liu & Li (’16), Phan et al. (’19), Simchowitz et al. (’21), Liu et al. (’22), Hamidi & Bayati (’23)]

Thompson sampling via autoregressive generation

Start with the outcomes we actually observed



Liu, Van Roy & Xu (’23) condition on generated future outcomes to guide exploration under known nonstationary dynamics

Misspecification

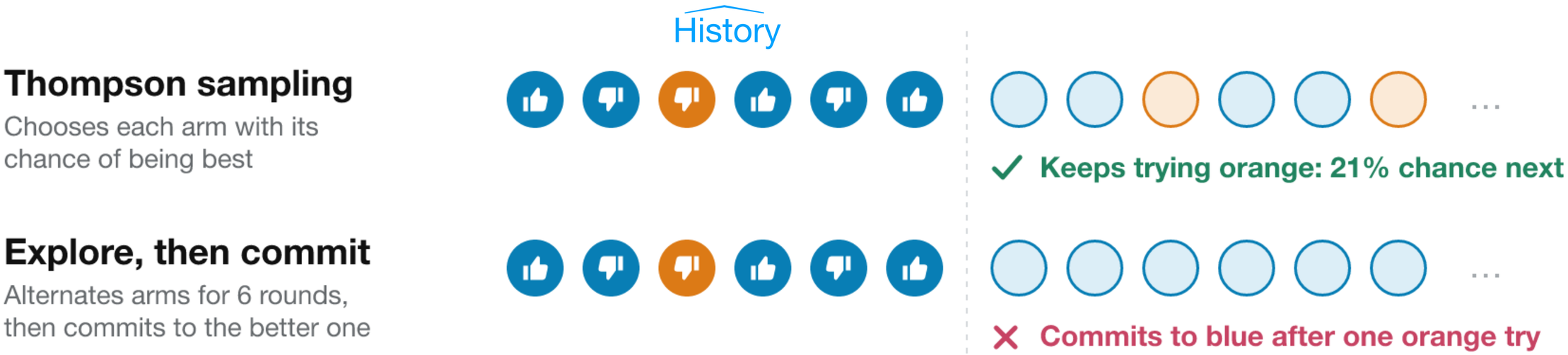
Can initially small errors cascade over time?

- $\pi^\star$: policy based on the true sequence model $p^\star$
  $\hat{\pi}$: policy based on the approximate sequence model $\hat{p}$



Is $\pi^\star$ any good on $\widehat{\pi}$'s history?

- Typical guarantees say $\pi^\star$ makes good decisions when it generates History*
- But what about $\pi^\star$'s decisions on $\overbrace{\text{History}}$?



Bayes regret controlled by sequence loss

- Assume potential outcomes across arms are independent
- Assume units/observations within an arm are exchangeable

Bayes regret controlled by sequence loss

- Assume potential outcomes across arms are independent
- Assume units/observations within an arm are exchangeable

Theorem (C**N**RZ'24)

Impossible to implement

Val loss

$$\text{Regret of TS with model } \widehat{p} \;\leq\; \text{Regret of oracle TS with true model } p^\star \;+\; \sqrt{2 \cdot \text{no. actions} \cdot (\ell_T(\widehat{p}) - \ell_T(p^\star))}$$

- On average, initially small disparities don't magnify over time
- Scaling laws govern online decision-making performance!

Back in 2019 at Meta



- Decisions have to be made fast in online platforms; TS too slow!

Theorem (**N**DB'26)	Imitating TS on the learner's own histories preserves the usual Bayes regret bound, up to **additive** imitation errors.

- Powered by the observation that TS is *robust to single-step errors*, we shipped a fast policy trained to mimic TS
- A/B test: up to 5× the share of videos preserved at 1080p, leading to 1.1% increase watch time.

Summary

- Meta-learning across datasets provides a robust prediction machine
- Our conceptual framework turns this into a new primitive for UQ: autoregressive generation is key to valid statistical inference!
- We formulate a MDP over predictive distributions, and approximately solve it using classical heuristics like TS and lookaheads
- For TS, approximation error accrues additively, and hence is controlled by sequence prediction loss

[Uncertainty as missing data] Active Exploration via Autoregressive Generation of Missing Data, C**N**RZ, 2024

[Language experiments] Adaptive Elicitation of Latent Information Using Natural Language, WZZ**N**, 2025

[Bayesian connections] Exchangeable Sequence Models Quantify Uncertainty Over Latent Concepts, Y**N**, 2024